

Course: Introduction to Indian Knowledge Systems

Course Code: S006

Credit:2

Course Description:

This course provides an overview of the Indian Knowledge Systems (IKS), which encompass the rich intellectual, cultural, philosophical and scientific traditions of ancient and classical India. Students will explore key aspects of philosophy, science, technology, arts, literature, and governance that originated and evolved in India over millennia. Through interdisciplinary study, the course highlights the relevance and application of these systems in the modern context, fostering an appreciation of India's intellectual heritage, its contributions to global knowledge and its relevance in addressing the problems faced by the today's world.

Note: Syllabus changes yearly and may be modified during the term itself, depending on the circumstances. Students, however, will be evaluated only on the basis of topics covered in the course.

Course Objectives:

- To introduce students to the foundational concepts, philosophies, and historical development of Indian Knowledge Systems.
- To explore key domains of IKS, such as Ayurveda, mathematics, astronomy, linguistics, and governance.
- To understand the interdisciplinary nature and practical applications of IKS in contemporary fields.
- To encourage critical analysis and appreciation of India's contributions to global thought and knowledge.
- To promote awareness of sustainable practices, ethical governance, and holistic well-being (physical, mental and spiritual) rooted in Indian traditions.

Course Outcomes: Upon successful completion of this course, students will be able to:

- Demonstrate a foundational understanding of the principles and philosophies underlying Indian Knowledge Systems.
- Identify and discuss key contributions of IKS in fields like science, medicine, mathematics, and arts.
- Analyse the relevance of IKS in addressing contemporary challenges in health, sustainability, and ethics.
- Appreciate the cultural and intellectual heritage of India with a critical perspective.
- Integrate principles from IKS into their personal and professional lives to foster holistic and sustainable practices.

Approximate weightage of different components in evaluation:

Attendance	40%
Assignment	15%
Final Exam	45%

Course Policies:

General

1. Electronic devices are not to be used during any exams unless instructed to do so.

Attendance and Absences

1. Attendance is expected and will be taken in each class. Students are not supposed to miss class without prior notice/permission. Any absences may result in point and/or grade deductions.
2. Students are responsible for all missed work, regardless of the reason for absence. It is also the absentee's responsibility to get all missing notes or materials.

Textbooks(s):

1. Introduction To Indian Knowledge System: Concepts And Applications, *B. Mahadevan, Vinayak Rajat Bhat, Nagendra Pavana R.N.*

Course Outline (tentative) and Syllabus:

The weekly coverage might change as it depends on the progress of the class. Each week assumes 1.5 hour lectures but additional classes may be taken (as and when required) to meet the credit requirement.

Week	Content
Week 1 - 2	• Indian Knowledge System - An Introduction • What is IKS & Why do we need IKS? • Some salient aspects and accomplishments of Traditional Knowledge Systems
Week 3 - 4	• The Vedic Corpus – Introduction and Classification, • Synopsis of Vedas, Vedāṅgas, Purāṇas, Itihāsas, Nīti-śāstras, and Subhāṣitas.
Week 5 - 6	• Mathematics – Indian Mathematicians and their Contributions • Unique aspects of Indian Mathematics
Week 7 - 8	• Medicine and Health Systems – Introduction to Ayurveda and its holistic approach • Charaka Samhita and Sushruta Samhita: Principles of diagnostics and surgery
Week 9 - 10	• Spirituality and Holistic Well-being
Week 11 - 12	• Indian Astronomy, • Pañcāṅga – The Indian calendar system, Astronomical Instruments (Yantras)
Week 13	• Engineering and Technology: Metals and Metalworking
Week 14	• Engineering and Technology: Other applications • Physical structures, Irrigation and water management, Dyes and painting technology, perfumery, Surgical techniques, Shipbuilding, Sixty-four art forms (64 Kalās)
Week 15 - 16	• Art, Literature & Architecture